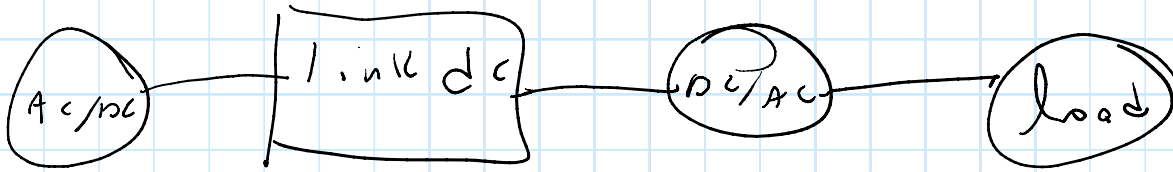


05-03-31

Sunday, June 21, 2026 8:51 AM

DC-AC Converters (inverters)

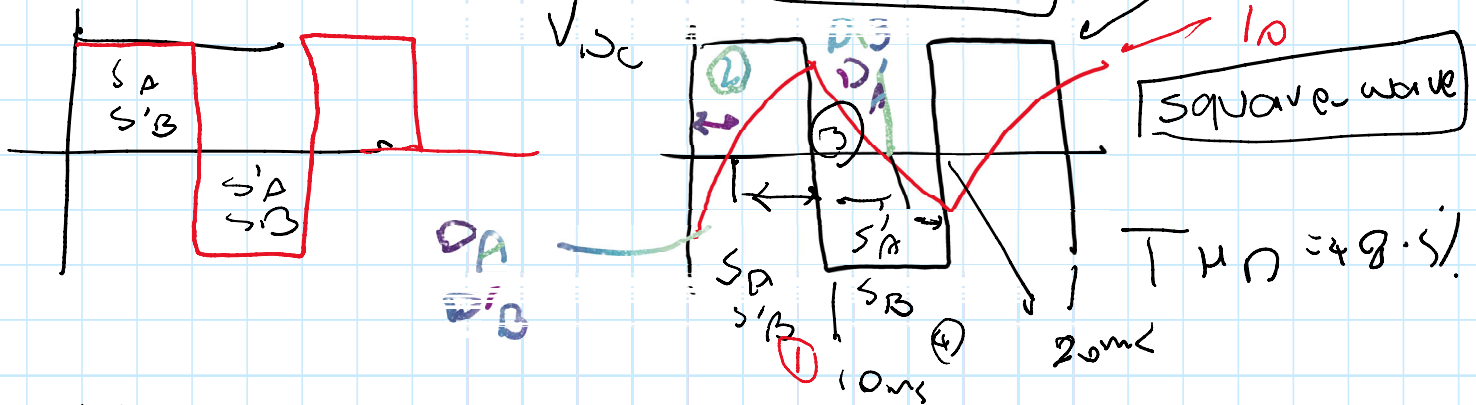
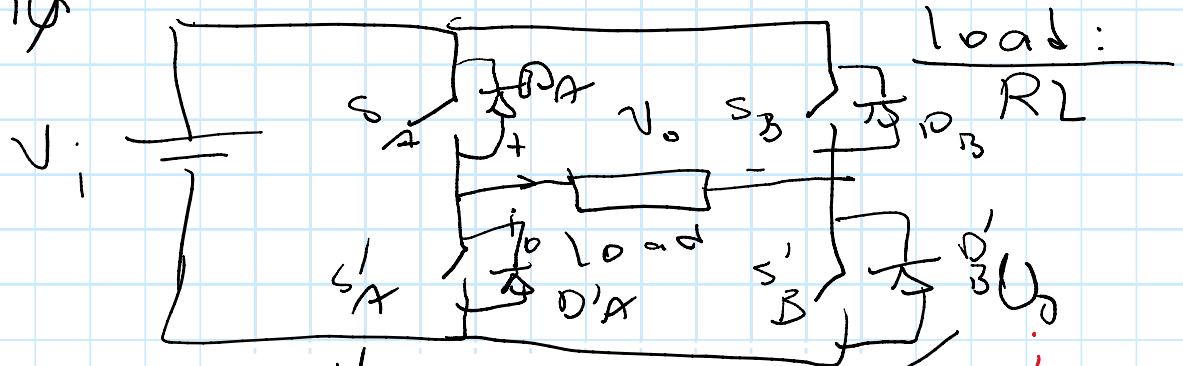
inverters $\begin{cases} 1\phi \\ 3\phi \end{cases}$ $\begin{cases} VSI - \text{Voltage Source Inv.} \\ CSI - \text{Current Source Inv.} \end{cases}$



VSI $\rightarrow$ Capacitor $\equiv$

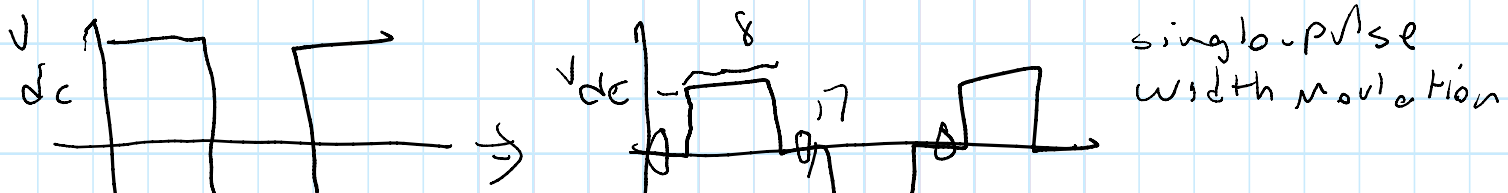
CSI $\rightarrow$ Inductor $\equiv$

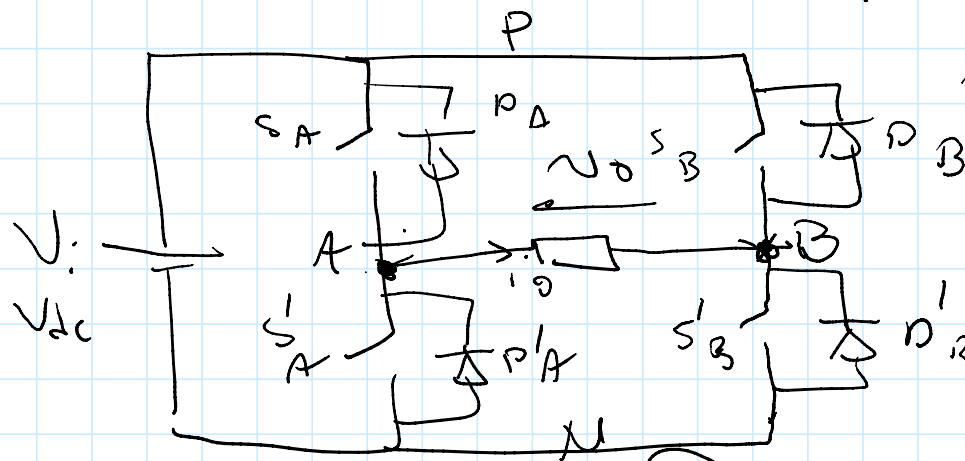
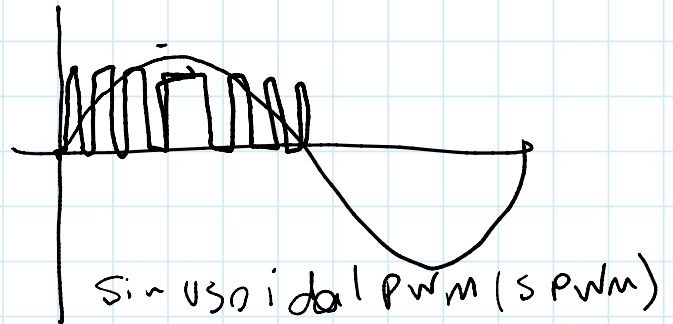
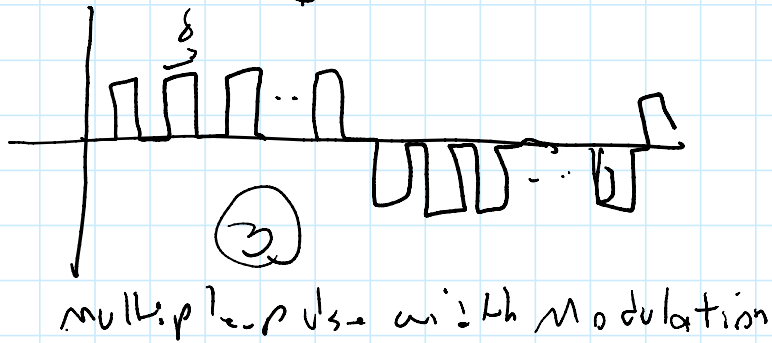
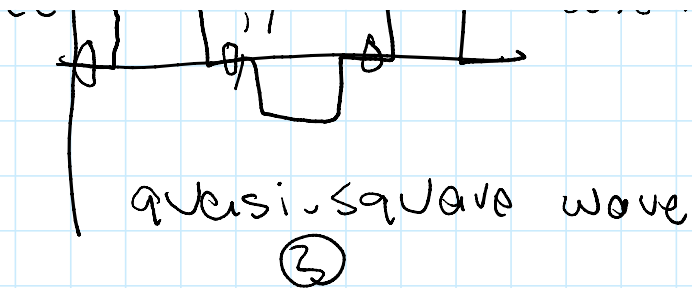
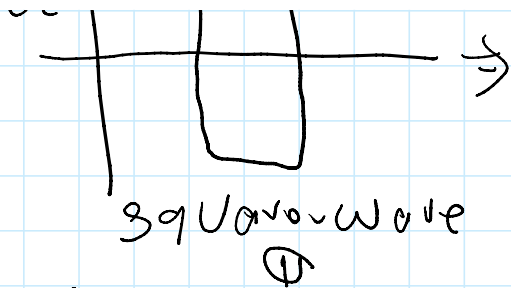
VSI - 1ϕ



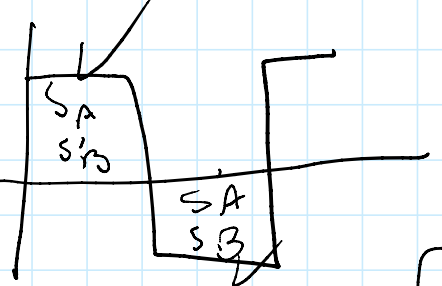
$$V_o = V_i = V_{dc}$$

PWM: Pulse-width modulation





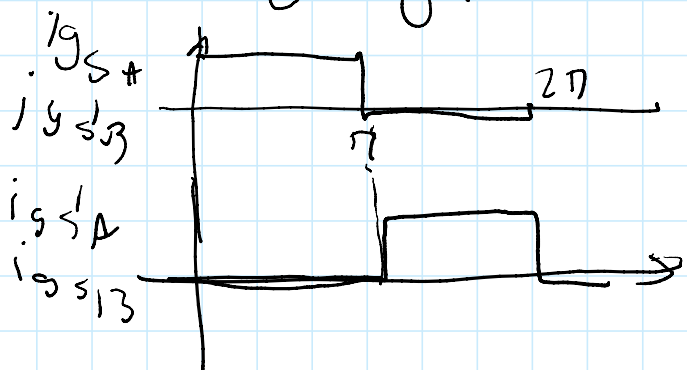
$$V_o = V_{AN} - V_{BN}$$



$$V_{AN} \begin{cases} S_A \text{ on } (S_{A'} \text{ off}) \rightarrow V_{AN} = V_i \\ S_A \text{ off } (S_{A'} \text{ on}) \rightarrow V_{AN} = 0 \end{cases}$$

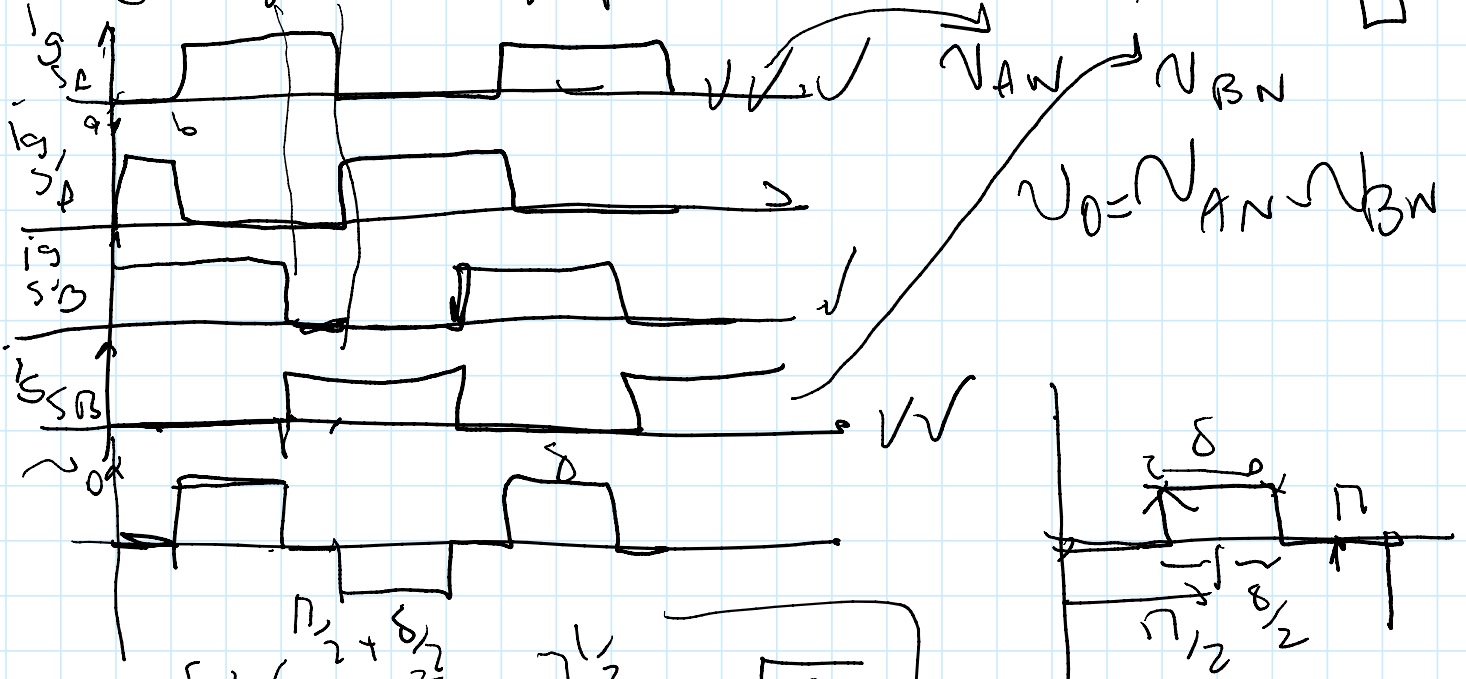
$$V_{BN} \begin{cases} S_B \text{ on } (S_{B'} \text{ off}) \rightarrow V_{BN} = V_i \\ S_B \text{ off } (S_{B'} \text{ on}) \rightarrow V_{BN} = 0 \end{cases}$$

switch gating pattern for square-wave operation.

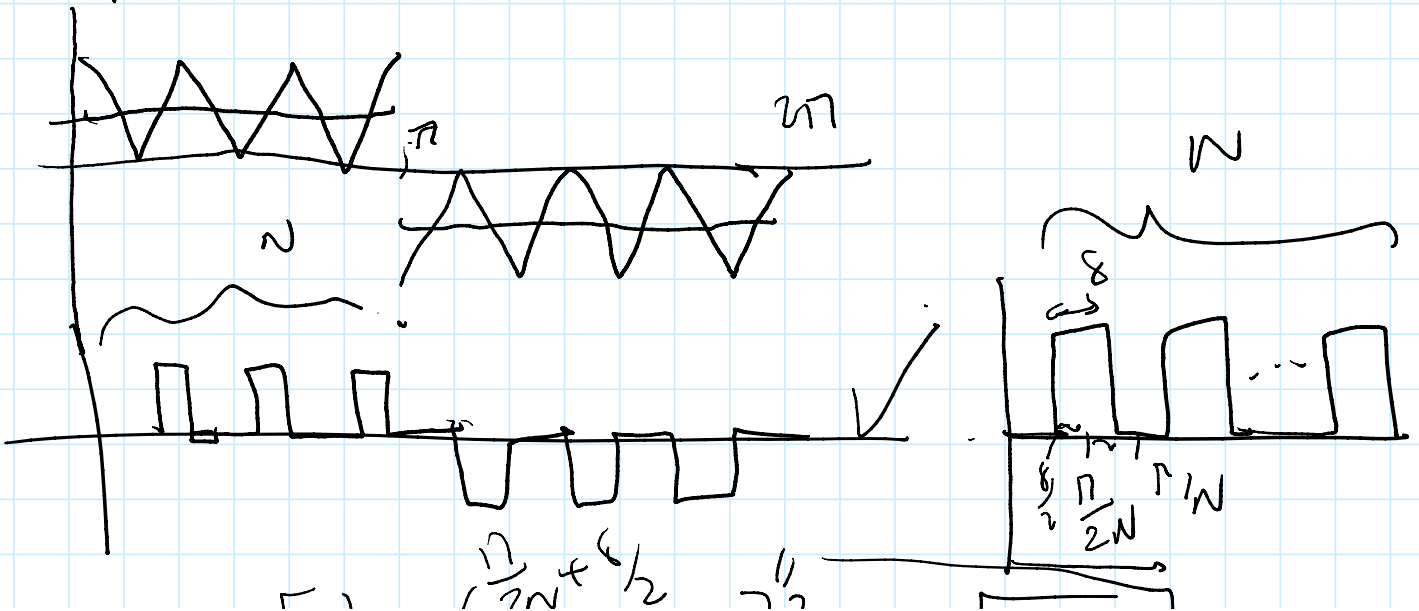
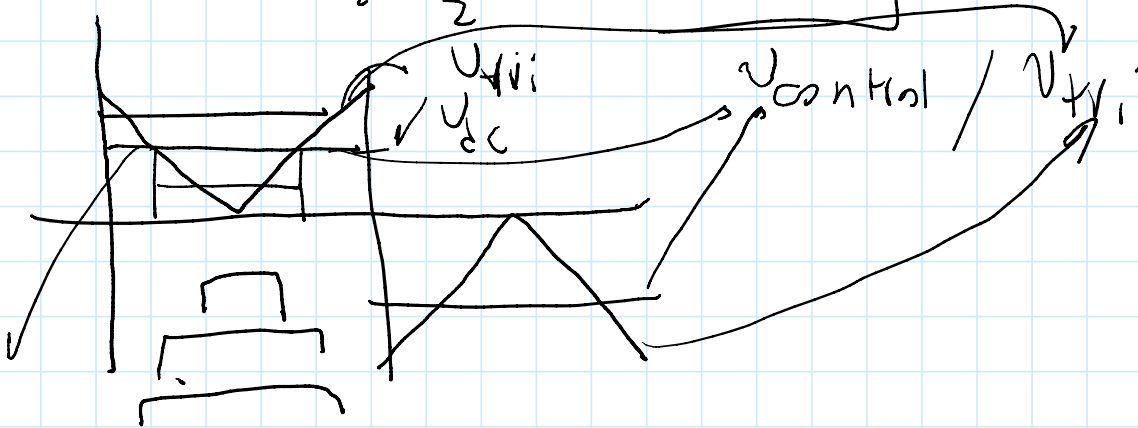
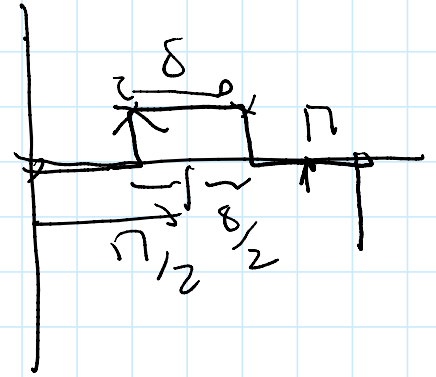


gatings for single-pulse width modulation.

Settings for single-pulse width modulation



$$V_o = \left[\frac{1}{\pi} \int_{\pi/2 - \delta/2}^{\pi/2 + \delta/2} V_m \sin \omega t d\omega t \right]^{1/2} = V_m \sqrt{\frac{\delta}{\pi}}$$

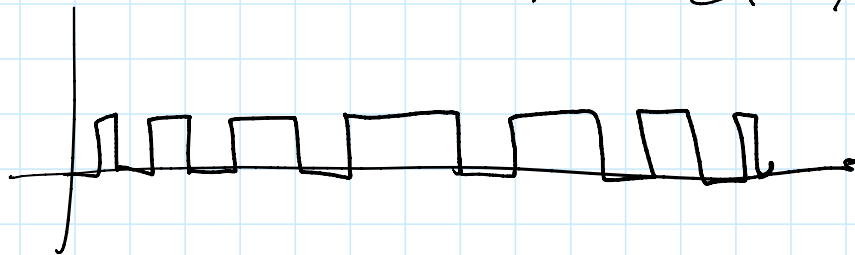


$$V_o = \left[\frac{1}{n/N} \int_{\frac{n}{2N} - \frac{\pi}{2}}^{\frac{n}{2N} + \frac{\pi}{2}} V_i^2 d\omega t \right]^{\frac{1}{2}} = V_i \sqrt{\frac{N\pi}{\pi}}$$

Sinusoidal PWM - SPWM

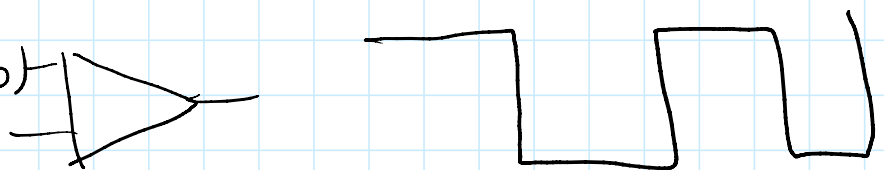
Unipolar

bipolar



Pulse Width Modulation - SPWM $V_o = V_{AN} - V_{BN}$

Carrier (triangle)
control



SPWM - Unipolar

Carrier - triangle

control - sinusoidal

S_A

if $V_{\text{control}} > V_{ti}$ — S_A ON (S'_A off)

↓
sinusoidal

otherwise <

— S_A off (S'_A on)

S_B

if $-V_{\text{control}} > V_{ti}$ — S_B ON (S'_B off)

otherwise < — S_B off (S'_B on)

m_m - V_{control}

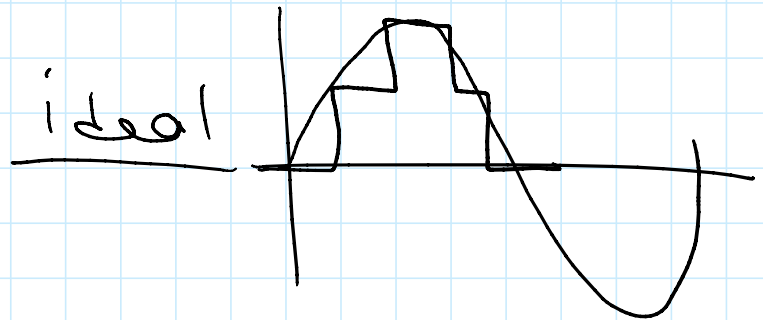
m₀ - carrier

$$M_a = \frac{V_{control}}{V_{tri}}$$

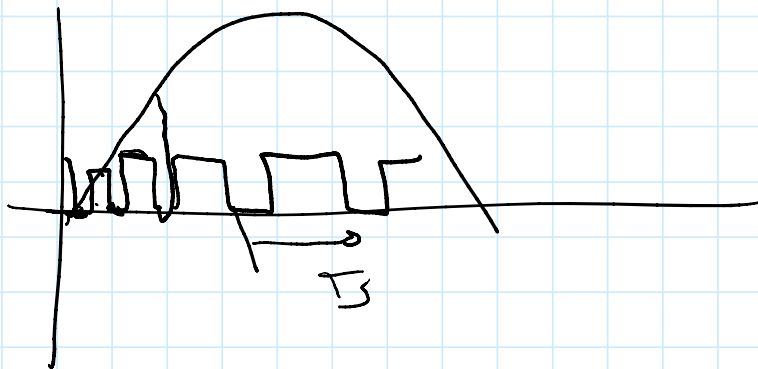
modulation index

$$m_f = \frac{f_{carrier}}{f_{control}}$$

frequency index



$N \rightarrow \infty$
 δ constant



local average
 average over a switching cycle

